

CardioTwin India — Digital Twin Architecture

SOURCE LAYER	FUSION / TWIN	INTELLIGENCE	CLINICIAN LAYER
Synthetic EHR Demographics Diagnoses Labs History	Digital Twin State Historical context + 7-day dynamic state	Feature Engineering + Logistic Regression + Explainability + Risk probability	Doctor Dashboard Risk band Trends Risk drivers Clinical review

Dynamic stream: heart rate • HRV • sleep • steps • activity • change from personal baseline

Core loop: Observe → Update Twin → Predict → Explain → Clinician Review

The PoC uses synthetic data only. Risk thresholds are prototype visualization thresholds and have not been clinically validated.